

## PHASE 8

# Project Demonstration Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

| Project Focus                                                                            |
|------------------------------------------------------------------------------------------|
| AI-assisted learning resources, secure backend services, and personalized study support. |

## Phase Objective

To present the completed AI StudyBuddy workflow and demonstrate how the backend, database, authentication, APIs, and AI services work together.

## 8.1 Demonstration Preparation

- Ensure MongoDB is available and the required environment variables are configured.
- Install project dependencies with npm install.
- Start the development server using npm start.
- Confirm that the backend connects to MongoDB before beginning API demonstrations.
- Use Postman or Thunder Client to send the demonstration requests.

## 8.2 Suggested Demonstration Flow

- Register a student account.
- Log in and obtain the JWT access token.
- Use the authenticated token for protected operations.
- Upload a study document or learning content.
- Request an AI-generated summary.
- Generate flashcards from the same material.
- Generate a multiple-choice quiz.
- Request a personalized study plan.
- Show how generated resources can be retained for later use.
- Demonstrate administrator-only operations where applicable.

## 8.3 Key Points to Explain During the Demo

- JWT authentication protects private API resources.
- MongoDB stores users, study material, and generated learning resources.
- Mongoose provides the schemas and validation used by the database layer.
- The Gemini service produces dynamic educational content rather than relying only on fixed templates.
- The modular backend separates routing, business logic, security, database access, and AI processing.

## 8.4 API Demonstration Endpoints

- POST /api/auth/register – create a user.
- POST /api/auth/login – authenticate and receive a token.
- POST /api/material/upload – submit study content.
- POST /api/materials/:id/summarize – create a concise summary.
- POST /api/ai/flashcards – create revision flashcards.
- POST /api/ai/quiz – generate assessment questions.
- POST /api/ai/study-plan – produce a personalized schedule.

## 8.5 Demonstration Evidence

The source documentation identifies separate backend-execution and API-testing demonstration video references. These can be used as supporting evidence alongside the live API walkthrough.

## **8.6 Final Demonstration Outcome**

The demonstration should show the complete path from user authentication and material submission to AI-generated learning resources, while also explaining how the backend architecture and database support the workflow.